ISA 444: Business Forecasting

02 - Introduction to Time Series Analysis

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Click here to schedule a virtual meeting

Spring 2022

Outline

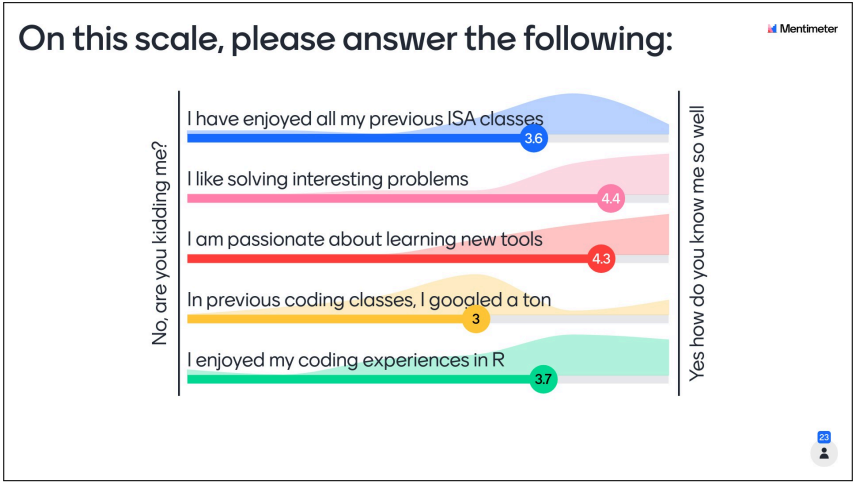
1. **Preface**
2. Types of Data Over Time
3. Components of a Time Series
4. Recap

Quick Refresher based on Last Class

Main Learning Outcomes

- ☑ **Describe course objectives & structure.**
- ☑ **Describe what do we mean by forecasting**
- ☒ **Explain the PIVASE framework**

Self-Assessed Skills from Last Class's Mentimeter Poll



Results from Last Class's Mentimeter Poll for Question A

Self-Described Goals for Taking the Class

More experience with predictive modeling, and an overall greater skillset for analyzing data

I'm hoping to gain tools to use in my future career and education in data analytics.

Hoping to expand my knowledge of forecasting and how it can be used in a variety of business environments.

I am hoping to refine the R skills that I currently have and gain new skills.

A more advanced understanding of how to write different queries, and understanding of their application and meaning.

Gain the tools needed to sufficiently use forecasting in a business context. I am a Finance major, however I believe forecasting/ISA principles are helpful in many fields.

I like being able to make data-driven decisions! I want to learn how to make strong predictions

learn more about R needed in the workplace

I hope to get a good understanding about how to take forecasting in the outside world and use it in my post grad life

I would like to build on the skills from ISA 291 so that I can become a better problem solver. I am taking this course as a thematic so I am hoping to gain additional skills to complement my ACC/FIN double major.

be able to forecast things, know more about forecasting tools and apply it to real-life problems

Learning more R programming skills for my graduate study.

to learn real knowledge that can be applied to real-life work about business forecasting

An understanding of how forecasting is done in business and how to avoid common pitfalls

I have always been interested in time series analysis and have heard great things about this class. I hope I am challenged in this class and come out of it with many new skills

I hope to improve my skills in model building and business analytics tools.

know how to deal with real world data and solve real world problem

I am hoping to improve my analytical skills to help advance my career.

Learning Objectives for Today's Class

Main Learning Outcomes

- **Explain the differences between cross sectional, time series, and panel datasets.**
- **Identify and describe the basic components of a time series including trends, seasonal components, and cycles.**

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1 Preface

2 Types of Data Over Time

3 Components of a Time Series

4 Recap

Cross Sectional Data [1]

Cross Sectional Data: Measurements on multiple units, recorded in a single time period.

Example 1: H1B 2020-2022 Data for Senior Data Scientists at Netflix[1]

START DATE	JOB TITLE	BASE SALARY	LOCATION
2021-08-11	SENIOR DATA SCIENTIST	118,955	LOS GATOS, CA
2021-06-14	SENIOR DATA SCIENTIST	143,291	LOS GATOS, CA
2021-06-14	SENIOR DATA SCIENTIST	143,291	LOS GATOS, CA
2021-09-09	SENIOR DATA SCIENTIST	143,291	LOS GATOS, CA
2021-10-02	SENIOR DATA SCIENTIST	149,261	LOS GATOS, CA
2021-07-31	SENIOR DATA SCIENTIST	198,203	LOS GATOS, CA
2020-11-29	SENIOR DATA SCIENTIST	360,000	LOS GATOS, CA
2020-10-01	SENIOR DATA SCIENTIST	375,000	LOS GATOS, CA
2020-05-19	SENIOR DATA SCIENTIST	400,000	LOS GATOS, CA
2020-10-02	SENIOR DATA SCIENTIST	410,000	LOS GATOS, CA

[1] Data scraped from https://h1bdata.info/index.php?em=Netflix+Inc&job=Senior+Data+Scientist&city=LOS+GATOS&year=All+Years on January 25, 2022 using the rvest package in R. The printing was limited to those individuals who started on/after January 01, 2020, with the filters specified in the URL.

Cross Sectional Data [2]

Cross Sectional Data: Measurements on multiple units, recorded in a single time period.

Example 2: Heart Disease Dataset[2]

age	sex	restingBP	maxHR	label
60.00	1.00	130.00	132.00	2
63.00	0.00	108.00	169.00	2
59.00	1.00	178.00	145.00	1
57.00	1.00	152.00	88.00	2
60.00	0.00	158.00	161.00	2
52.00	1.00	125.00	168.00	2
45.00	1.00	128.00	170.00	1
51.00	1.00	140.00	122.00	2
58.00	1.00	140.00	165.00	1
51.00	1.00	100.00	143.00	1

[2] Data sampled from this UCI Machine Learning Repository.

Cross Sectional Data [3]

Cross Sectional Data: Measurements on multiple units, recorded in a single time period.

Example 3: NBA 2021-2022 Leaders - Top 10 in PTS/Game[3]

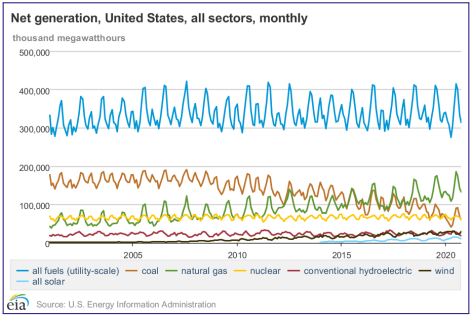
Player	Pos	Age	Tm	G	FG	FG%	eFG%	PTS
Kevin Durant	PF	33	BRK	36	10.5	.520	.566	29.30
LeBron James	SF	37	LAL	35	10.8	.518	.585	29.00
Joel Embiid	C	27	PHI	35	9.3	.498	.534	28.70
Giannis Antetokounmpo	PF	27	MIL	39	9.9	.531	.560	28.60
Trae Young	PG	23	ATL	42	9.3	.455	.526	28.00
DeMar DeRozan	PF	32	CHI	42	9.5	.495	.511	26.30
Nikola Jokic	C	26	DEN	40	10.2	.573	.617	26.10
Stephen Curry	PG	33	GSW	43	8.4	.416	.535	26.00
Jayson Tatum	SF	23	BOS	44	8.9	.422	.486	25.80
Luka Doncic	PG	22	DAL	32	9.2	.437	.492	25.60

[3] Data scraped from Basketball-Reference on January 25, 2022 using the rvest package in R. The printing was limited to the top 10 players and the selected variables.

Time Series Data [1]

Time Series Data: Comparable measurements recorded on a single (or a few) variables over time (usually a long period of time).

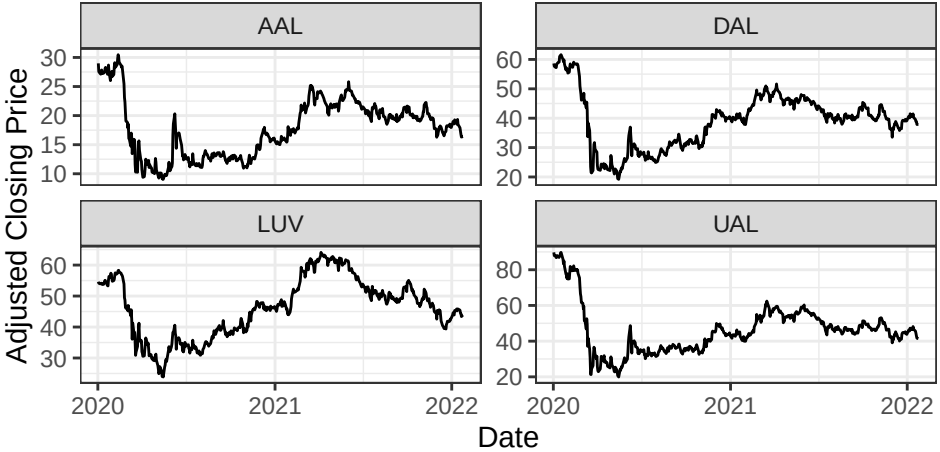
Example 1: Net Power Generation in the U.S.



Net monthly electricity generation in the U.S. from the U.S. Energy Information Adminstration.

Time Series Data [2]

Example 2: Stock prices of U.S. Airlines



As of January 25, 2022: COVID-19 had a substantial impact on these airline stocks.

Panel Data

Panel Data: Cross sectional measurements (usually many variables) repeated over time (usually over a few time periods).

Example: World Bank's Data[4]

iso3c	date	NY.GDP.MKTP.KD.ZG	SH.DYN.NMRT	SH.HIV.INCD.ZS	SH.MED.BEDS.ZS	SH.MED.PHYS.ZS	SH
CHN	2018.00	6.75	4.20		4.31	1.98	
CHN	2019.00	5.95	3.90		4.31	1.98	
CHN	2020.00	2.35	3.90		4.31	1.98	
EGY	2018.00	5.31	11.50	0.05	1.43	0.77	
EGY	2019.00	5.56	11.10	0.05	1.43	0.75	
EGY	2020.00	3.57	11.10	0.06	1.43	0.75	
USA	2018.00	3.00	3.80	0.21	2.87	2.60	
USA	2019.00	2.16	3.70	0.20	2.87	2.60	
USA	2020.00	-3.64	3.70	0.20	2.87	2.60	

[4] Data queried from the World Bank Data Catalog using the wbstats package in R. The printed results show a snapshot of 7 variables (out of a much larger panel dataset). You can think of panel data as a cross-sectional dataset with a longitudinal/time component.

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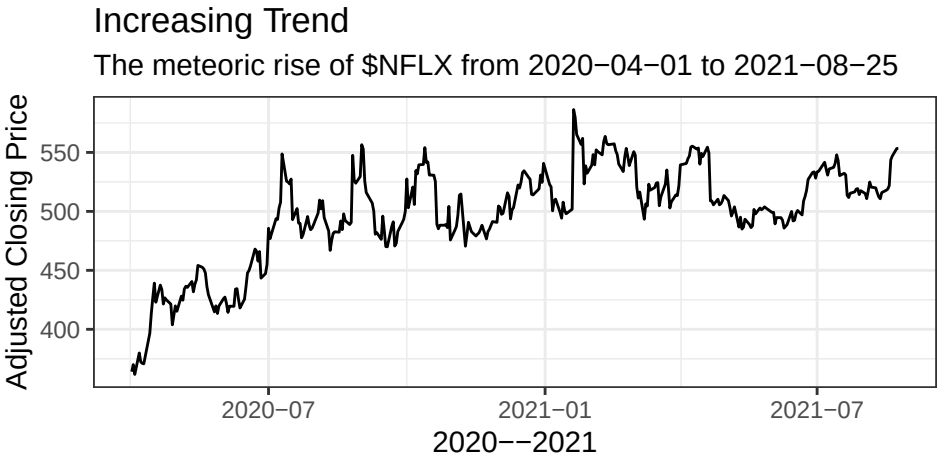
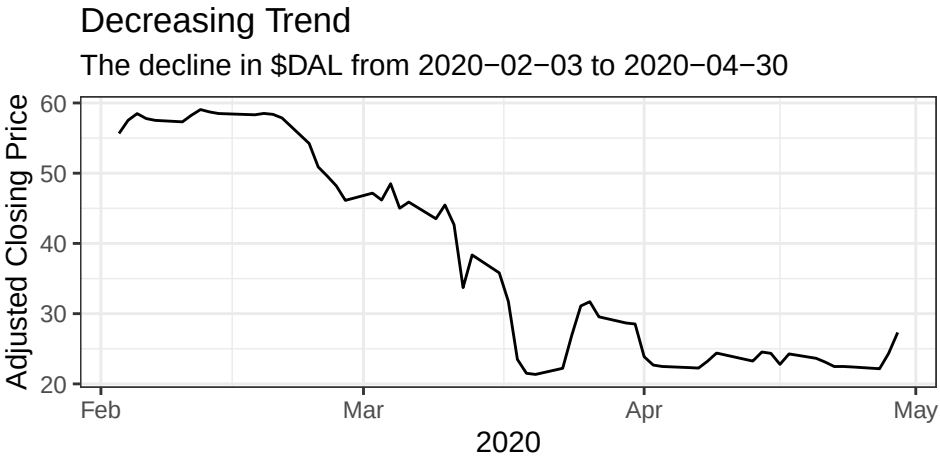
Trend [1]

A **trend** is an increasing or decreasing pattern over time.



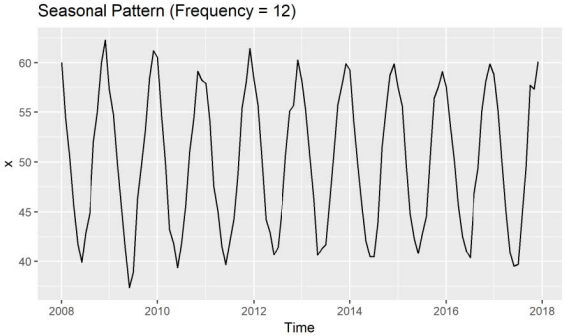
Trend [2]

A **trend** is an increasing or decreasing pattern over time.



Seasonality [1]

Seasonality refers to the property of a time series that displays REGULAR patterns that repeat at a constant frequency (m).

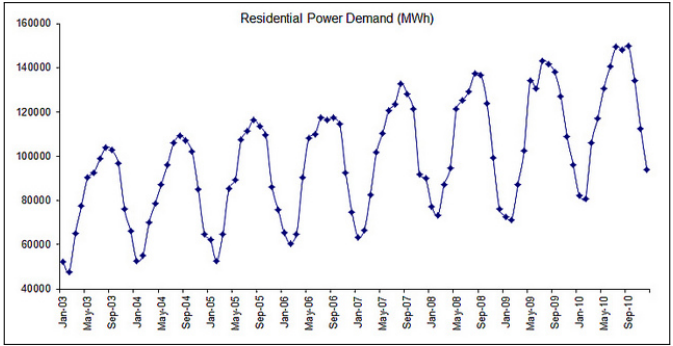


A time series with a monthly seasonal pattern.[6]

[6] Figure is from Dr. Allison Jones-Farmer's lecture notes, Miami University, Spring 2020.

Seasonality [2]

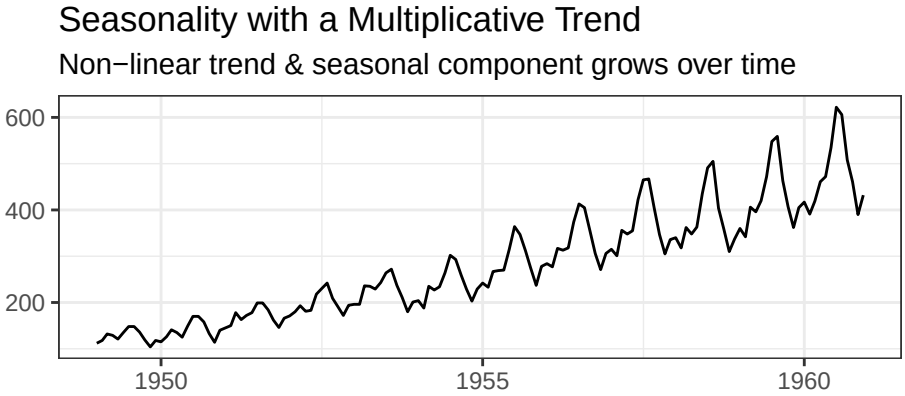
Seasonality refers to the property of a time series that displays REGULAR patterns that repeat at a constant frequency (m).



Seasonality with an Additive Trend[7].

[7] Figure is from Patterns Unplugged - Help Center published online on December 28, 2016.

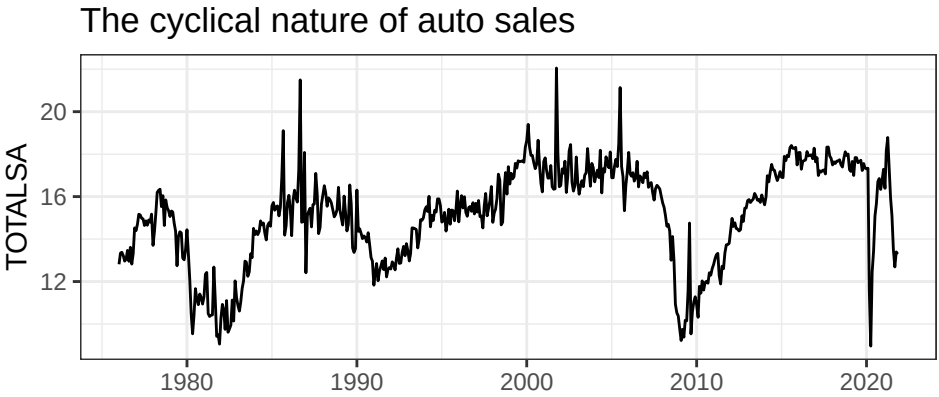
Seasonality [3]



AirPassengers R Dataset −− Source: Box, G. E. P., Jenkins, G. M. and Reinsel, G. C. (1976) Time Series Analysis, Forecasting and Control.

Cycle [1]

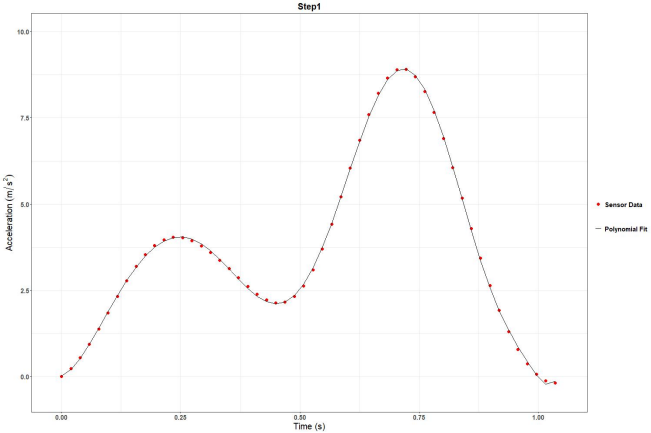
Cyclical fluctuations are somewhat irregular (unknown duration).



Total Vehicle Sales [TOTALSA], retrieved from FRED, Federal Reserve Bank of St. Louis https://fred.stlouisfed.org/series/TOTALSA, January 25, 2022.

Cycle [2]

Cyclical fluctuations are somewhat irregular (unknown duration).



A gait "cycle" from an IMU attached to the ankle. Joint work with the University at Buffalo.

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Summary of Main Points

Main Learning Outcomes

- **Explain the PIVASE framework**
- **Explain the differences between cross sectional, time series, and panel datasets.**
- **Identify and describe the basic components of a time series including trends, seasonal components, and cycles.**

Things to Do

- Thoroughly read Chapter 1 of our reference book, which can be downloaded from the Publisher.
- Go through the slides, examples and make sure you have a good understanding of what we have covered.
- **Optional, but highly recommended,** go through the self-paced interactive review for Week 01.
- **Required:** Complete the graded assignment.

Self-Paced Interactive Review for Week 01

To help you brush your `R` skills and assess your understanding of the material covered in Week 01, I have created a self-paced interactive review. The review can be accessed at http://rstudio.fsb.miamioh.edu:3838/megahefm/isa444/week01/.

This review is optional, but if you plan on completing it, you are advised to do the following prior to going through the self-paced review.

1. Read Chapter 01 from our textbook.
2. Have completed the in-class examples in the lecture notes for Week 01.
3. Be able to dedicate about 20-40 minutes to go through this entire tutorial.

Graded Assignment 02: Evaluating your Understanding

Please go to Canvas (click here) and answer the four questions. **Due August 31, 2021 [08:30 AM].**

What/Why/Prep? The purpose of this assignment is to evaluate your understanding and retention of the material covered up to the End of Class 02. Furthermore, you should have thoroughly read Chapter 01 of the book.

General Guidelines:

- Individual assignment.
- This is **NOT** a timed assignment. If the concepts we covered are well-understood, this should take 10-15 minutes.
- Proctorio is NOT required for this assignment.
- You should have access to RStudio.

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